ETALE SEPARATED MORPHISMS COMMUTE WITH THE KEEL-MORI MODULI SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4107-4129 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4131-4189, the complete author proof (57 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 23 of queue_stacks-stacks-more-morphisms.json (pb=2178, stmt 4107-4129, proof 4131-4189).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-etale-local-finite-inertia: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3798–3814. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-well-nigh-affine-moduli-space: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3425–3433. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DU9: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4189–4189. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DTW: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4189–4189. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma theorem-keel-mori: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3854–3862. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $h : \mathcal{X}' \to \mathcal{X}$ be a morphism of algebraic stacks. Assume*

(1) *$\mathcal{I}_\mathcal{X} \to \mathcal{X}$ is finite,*
(2) *h is étale, separated, and induces isomorphisms on automorphism groups (Morphisms of Stacks, Remark 0DTW).*

Then there exists a cartesian diagram

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ M' & \longrightarrow & M \end{array}$$

where $M' \to M$ is a separated étale morphism of algebraic spaces and the vertical arrows are the moduli spaces constructed in Theorem theorem-keel-mori.

Proof. By Morphisms of Stacks, Lemma 0DU9 we see that $\mathcal{I}_{\mathcal{X}'} \to \mathcal{X}' \times_\mathcal{X} \mathcal{I}_\mathcal{X}$ is an isomorphism. Hence $\mathcal{I}_{\mathcal{X}'} \to \mathcal{X}'$ is finite as a base change of $\mathcal{I}_\mathcal{X} \to \mathcal{X}$. Let $f' : \mathcal{X}' \to M'$ and $f : \mathcal{X} \to M$ be as in Theorem theorem-keel-mori. We obtain a commutative diagram as in the lemma because f' is categorical moduli space. Choose I and $g'_i : \mathcal{X}'_i \to \mathcal{X}'$ as in Lemma lemma-etale-local-finite-inertia. Observe that $g_i = h \circ g'_i$ is étale, separated, and induces isomorphisms on automorphism groups (Morphisms of Stacks, Remark 0DTW). Let $f'_i : \mathcal{X}'_i \to M'_i$ be as in Lemma lemma-well-nigh-affine-moduli-space. In the proof of Theorem theorem-keel-mori we have seen that the diagrams

$$\begin{array}{ccc} \mathcal{X}'_i & \xrightarrow[g'_i]{} & \mathcal{X}' \\ {\scriptstyle f'_i}\downarrow & & \downarrow{\scriptstyle f'} \\ M'_i & \longrightarrow & M' \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathcal{X}'_i & \xrightarrow[g_i]{} & \mathcal{X} \\ {\scriptstyle f'_i}\downarrow & & \downarrow{\scriptstyle f} \\ M'_i & \longrightarrow & M \end{array}$$

are cartesian and that $M'_i \to M'$ and $M'_i \to M$ are étale (this also follows directly from the properties of the morphisms g'_i, g_i, f', f'_i, f listed sofar by arguing in exactly the same way). This first implies that $M' \to M$ is étale and second that the diagram in the lemma is cartesian. We still need to show that $M' \to M$ is separated. To do this we contemplate the diagram

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X}' \times_\mathcal{X} \mathcal{X}' \\ \downarrow & & \downarrow \\ M' & \longrightarrow & M' \times_M M' \end{array}$$

The top horizontal arrow is universally closed as $\mathcal{X}' \to \mathcal{X}$ is separated. The vertical arrows are as in Theorem theorem-keel-mori (as flat base changes of $\mathcal{X} \to M$) hence universal homeomorphisms. Thus the lower horizontal arrow is universally closed. This (combined with it being an étale monomorphism of algebraic spaces) proves it is a closed immersion as desired. □